

Scales : 4/16/26

Explaining the KL divergence

I find it helpful to think about the KL divergence in terms of student S and teacher T distributions. By construction, we always optimize the student.

$$\textcircled{1} \quad D_{KL}(S \parallel T) = \mathbb{E}_S \log \frac{S}{T}$$

We sample from student, this blows up if T is small.

So this is mode seeking, since everywhere we are likely to sample must have some density in T .

$$\textcircled{2} \quad D_{KL}(T \parallel S) = \mathbb{E}_T \log \frac{T}{S}$$

We sample from teacher. We need S to not be small for any sample, so this is "covering". We want to have "some" mass anywhere we might sample from T , even if we don't get the modes exactly right.